

Visualizing diffusion posterior density for seismic inversion

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SUMMARY

Diffusion posterior sampling (DPS) combines Bayesian inversion with a learned implicit prior to generate high-fidelity subsurface models. Existing assessments of DPS often rely on scalar metrics. Further, recent statistical benchmarks are limited to linear inverse problems thereby limiting our intuition about why the method fails under challenging inversion conditions. In this work we augment such metrics with qualitative interpretation of the learned posterior to provide better insight into when these methods succeed or fail. This paper proposes a qualitative diagnostic to evaluate DPS by visualizing a low-dimensional projection of the posterior density. This visualization reveals features of the posterior geometry that predict and explain performance degradation due to band-limited data and out-of-distribution geological structures.

INTRODUCTION

Seismic inversion is ill-posed because multiple (including geologically implausible) subsurface models can explain the measured data (Tarantola, 1987; Symes et al., 2025), making uncertainty estimation essential. Bayesian inversion allows for quantification of this uncertainty by characterizing the posterior density over models conditioned on measured waveforms (Duijndam, 1988). However, parametric priors fail to capture the complex distribution of realistic subsurface models. (Gouveia and Scales, 1998; Alfayez et al., 2025). Diffusion models circumvent the need for explicit prior parametrization by learning an implicit prior (Ho et al., 2020). Diffusion posterior sampling (DPS) combines this learned prior with data misfit information to sample from the posterior density (Daras et al., 2024; Chung et al., 2023; Taufik and Alkhalifah, 2026; Cirakman et al., 2025).

Despite these advantages, qualitative evaluation of the resulting diffusion-based posterior density has yet to be extensively analyzed. Such diagnostics are important because scalar metrics and single model reconstructions do not reveal posterior structure (Zach et al., 2026). We propose a qualitative diagnostic for posterior evaluation by visualizing the low-dimensional projected posterior density and demonstrate that this diagnostic effectively predicts DPS performance. Since the posterior is in high-dimensional model space, direct visualization is impossible. Following the work by Yair et al. (2024) and Izzatullah et al. (2019), we visualize a low-dimensional representation of the posterior by projecting an ensemble of samples onto the plane defined by the dominant variance directions, i.e., the principal components of the centered sample matrix (Izzatullah et al., 2021; Jolliffe, 1986). Visualizing the kernel density estimate within this plane provides an interpretable view of the posterior density (Silverman, 2018). We explore the posterior for synthetic experiments (with known ground-truth models)

and find that features of this visualization effectively characterize reconstruction quality.

The outline of this paper is as follows. First, we discuss seismic inversion using DPS and then describe the theoretical framework for visualizing the posterior landscape. Our numerical experiments demonstrate that alignment of the posterior mode with the ground-truth predicts accuracy of DPS-based inversion, and the concentration of samples predicts average model uncertainty. This visualization clearly shows that DPS performance degrades when using band-limited data (analogous to cycle-skipping in deterministic inversion) and when the ground-truth model is out-of-distribution relative to the training data.

THEORY

Bayesian inversion

In this work we assume that wave propagation is modeled by the 2D constant-density acoustic wave equation (Alford et al., 1974). We solve a Bayesian inverse problem to determine the posterior density over sound velocity models by updating the prior (a measure of geological realism traditionally independent of measured data) together with the likelihood (enforcing consistency between the measured and simulated waveforms), i.e.,

$$P(m|d) \propto P(d|m)P(m). \quad (1)$$

Here d denotes the measured data, m the model parameters, and F is our forward model (the acoustic wave operator). The posterior density is $P(m|d)$, and $P(m)$ is our prior (Grether, 1980). Since seismic data errors arise from independent random perturbations, e.g. due to ambient noise or instrumentation malfunctions, we can model this error as additive Gaussian noise (Zhong et al., 2015). Thus we assume the likelihood $P(d|m)$ is Gaussian:

$$P(d|m) \propto e^{-\|d-F(m)\|^2}. \quad (2)$$

Accuracy of the prior determines posterior sample quality (Bosch et al., 2010). However, parametric priors often fail to capture complex model distributions common in realistic settings. Diffusion models learn an implicit prior from a training set of velocity models, thereby removing the need for explicit prior parametrization (Alfayez et al., 2025).

Diffusion prior

Diffusion models learn an implicit prior over realistic subsurface models by training a neural network to remove added noise. In the denoising diffusion probabilistic model (DDPM) (Ho et al., 2020), the forward process gradually adds noise to clean samples, $m_0 \sim P(m)$, in a series of time steps, generating noisy latent variables $\{m_t\}_{t=1}^T$. Training minimizes the error between the added noise and the predicted noise, $\epsilon_\theta(m_t, t)$,

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by updating the network parameters, θ . The reverse process uses these predictions to denoise latent variables and generate samples from the prior.

Intuitively, denoising corresponds to shifting the latent variables to high-probability regions of the noisy latent density $P_t(m_t)$. The score, i.e. the gradient of log-density, also describes the same direction. Song and Ermon (2019) show that the predicted noise is proportional to the negative score of the noisy latent density, i.e. $\varepsilon_\theta(m_t, t) \propto -\nabla_{m_t} \log P_t(m_t)$. Posterior sampling requires augmentation of the denoising process with the likelihood score in the same latent space, which is intractable to compute (Daras et al., 2024).

Diffusion posterior sampling

In DPS, we use an approximation proposed by Chung et al. (2023), which replaces the intractable latent likelihood score with the misfit gradient. Specifically, we approximate the likelihood score with respect to the noisy latent variable m_t by computing the data-misfit gradient at the denoised estimate $\hat{m}_0$ recovered from m_t , i.e.,

$$\nabla_{m_t} \log P_t(d|m_t) \approx -\nabla_{m_t} \|d - F(\hat{m}_0)\|^2. \quad (3)$$

Thus, we define the posterior score of the noisy latent density as,

$$\nabla_{m_t} \log P_t(m_t|d) \approx -\varepsilon_\theta(m_t, t) - \nabla_{m_t} \|d - F(\hat{m}_0)\|^2. \quad (4)$$

The first term in (4) enforces geological plausibility and the second term enforces data consistency of the latent variable. Denoising along this posterior score generates samples from the approximate posterior density, yielding the diffusion posterior sampling (DPS) method.

Visualization of posterior landscape

Density estimation directly from generated samples is infeasible due to the high dimensionality of the model space (Silverman, 2018). Instead we project the samples onto a 2D plane along the directions of maximal variance. We find these directions from the first two principal components, d_1 and d_2 , of the centered sample matrix

$$V = [x_1 - \bar{x}, x_2 - \bar{x}, \dots, x_n - \bar{x}] \quad (5)$$

where $\{x_i\}_{i=1}^n$ denote posterior samples consistent with data d and $\bar{x}$ denotes the sample mean (Jolliffe, 1986). Projecting onto this subspace preserves the highest amount of posterior variability, thereby providing a meaningful visualization of the posterior landscape.

Since we have no explicit formulation of the projected posterior density, we use kernel density estimation (KDE) to build a smooth nonparametric surrogate of the projected marginal posterior density (PMPD) from the projected samples (Silverman, 2018). The KDE yields a function approximation of the PMPD by averaging Gaussian densities centered at the projected samples. Next, we parametrize the principal subspace relative to the projected sample mean and use the basis $\{d_1, d_2\}$. This parametrization enables us to visualize the PMPD from the contour plot of

$$f(\alpha_1, \alpha_2) = \hat{q}(\Pi(\bar{x}) + \alpha_1 d_1 + \alpha_2 d_2), \quad (\alpha_1, \alpha_2) \in \mathbb{R}^2. \quad (6)$$

where $\hat{q}$ is the KDE of the PMPD and Π denotes projection onto the dominant variance plane.

NUMERICAL RESULTS

This section documents results from two experiments designed to visualize the PMPD. In both experiments, we perform seismic inversion of a synthetic subsurface model using DPS. We use the same learned diffusion prior and acquisition geometry for both sets of experiments. Our diffusion prior is a 2D DDPM model (U-net architecture) trained on 5000 velocity patches, each of size 512 m in x and z , uniformly sampled from the Marmousi velocity model (Brougois et al., 1990). We conduct all experiments in PyTorch using the NVIDIA A100 GPU runtime on Google Colaboratory (Paszke et al., 2019; Bisong, 2019).

We generate the observed data by simulating wave propagation through the Marmousi velocity model using the Deep-wave library (Richardson, 2026). The acquisition includes 16 isotropic point sources in space convolved with a 7 Hz Ricker wavelet in time. We record the seismic responses using 32 receivers, each recording for 1.4 seconds with a 4 millisecond sampling interval.

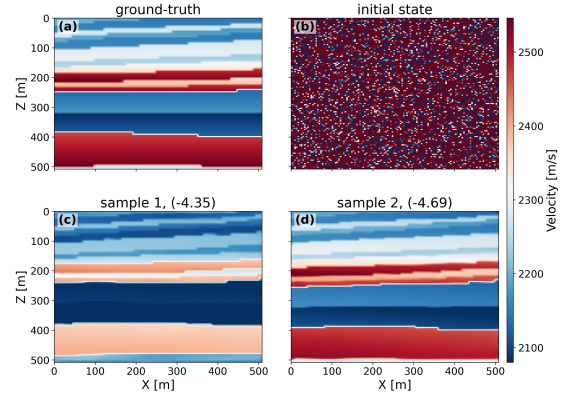


Figure 1: Experiment 1a plots: (a) The ground truth model for Experiment 1. (b) An example of the pure noise initial state. (c,d) Two (out of 50) samples generated by DPS-based inversion from full-bandwidth data. The log-misfit of the posterior samples are shown in parenthesis next to the sample number.

In each experiment we perform inversion using DPS initialized from pure noise and generate 50 posterior samples. Next, we visualize the PMPD from the contour plots of the KDE of projected samples. The plots display concentration of the posterior samples, describing estimation uncertainty, and multimodality. We also overlay these plots with data misfit contours. We find that alignment of the ground-truth with the posterior modes and the misfit minima qualitatively indicate accurate DPS inversion. Experiment 1 explores the impact of band-limited data on the posterior density, while Experiment 2 studies the posterior for an out-of-distribution model.

Experiment 1 (inversion on data without low-frequencies)

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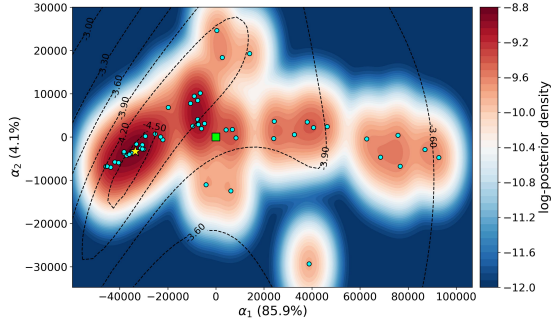


Figure 2: The PMPD plot corresponding to full-bandwidth data centered around the sample mean (green square) for Experiment 1a. The posterior samples are shown as cyan circles, and the ground truth model is shown as a yellow star. The black dashed lines show the log-misfit contours. The percent variation of the samples along each principal component (α_1 and α_2) are shown in parenthesis on the axes.

Experiment 1 investigates the effect of missing low-frequency content in the data on the DPS posterior landscape. We use a subset of the Marmousi model as the ground-truth (see Figure 1a). In Experiment 1a, we perform DPS-based inversion from full-bandwidth data and in Experiment 1b we use data lacking sub 5 Hz frequencies.

Experiment 1a: full-bandwidth data

This experiment demonstrates DPS performance under ideal inversion conditions where the data contains all frequencies. Figure 1 shows the ground-truth model, the noisy initial state, and two representative posterior samples. Figure 2 shows the corresponding posterior landscape in the principal component plane.

From Figure 1, we find that the posterior samples accurately recover the layered structure of the ground-truth with some samples exhibiting incorrect velocity ranges. The PMPD plot (Figure 2) is multimodal with two posterior modes: 1) near the ground-truth and 2) near the sample mean. The alignment of the misfit minima with the posterior mode near the ground-truth demonstrates that with full-bandwidth data, DPS successfully guides the sampling process to data-consistent and geologically-plausible models. Strong concentration of samples near the posterior modes also suggests low estimation uncertainty.

Experiment 1b: data lacking low-frequency content

In this experiment, we are interested in reproducing real-world seismic data acquisition conditions where low-frequency data is challenging to record due to either receiver limitations or noise contamination. Thus, we remove sub 5 Hz content in the data using a smooth filter and perform DPS-based inversion on this band-limited data. Figure 3 displays the ground-truth velocity model, a noisy initial state, and two representative posterior samples. Figure 4 shows the estimated posterior landscape in the principal component plane. We note that the misfit contours in Figure 4 show the high-frequency (> 5 Hz) data misfit.

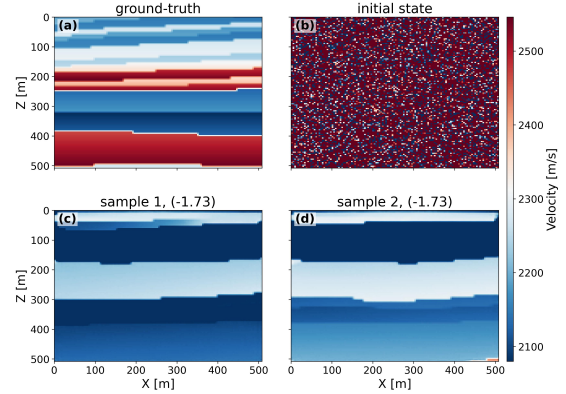


Figure 3: Experiment 1b plots: (a) The ground truth model for Experiment 1. (b) An example of the pure noise initial state. (c,d) Two (out of 50) samples generated by DPS-based inversion from data lacking sub 5 Hz frequencies. The log-misfit of the posterior samples are shown in parenthesis next to the sample number.

From Figure 3, we find that both posterior samples recover sections of the shallow layered structure but show incorrect velocity ranges. Comparing against samples in Figure 1 makes clear the necessity of low-frequency data for constraining the background velocity. From Figure 4 we see that the samples settle in a local minimizer of the log-misfit instead of the true minimum (low-misfit basin around the ground-truth) as in Figure 2. The posterior mode aligns with this local minimum, and the ground-truth lies near the edge of the posterior support. Since the principal components capture 90% of the sample variation, the displacement of the ground-truth from the posterior modes is unlikely to be a projection artifact. The strong concentration of the samples near the posterior modes suggests low uncertainty of the incorrect subsurface models, i.e. erroneous model estimation with high certainty. This observation indicates that DPS-based inversion is susceptible to cycle-skipping.

Experiment 2 (out-of-distribution ground-truth)

In this experiment, we consider as our ground-truth model a velocity model from the CurveFault-B class of the OpenFWI dataset (Deng et al., 2022). This model is out-of-distribution (OOD) relative to the training data. Figure 5 displays the ground truth, an example of the initial state, and two (out of 50) posterior samples generated using full-bandwidth data. Figure 6 shows the corresponding PMPD plot in the principal component plane.

From Figures 5c,d we find that the reconstructed velocity profiles are biased towards smooth structures and a reduced velocity range compared to the faulted structure in the ground-truth. The diffusion prior is biasing the denoising steps towards in-distribution geology. From Figure 6, we find that the ground-truth lies outside the posterior support and since the principal components capture 84.3% of the sample variation, this separation is unlikely to be a projection artifact. The displacement of the ground-truth from the posterior support indicates the

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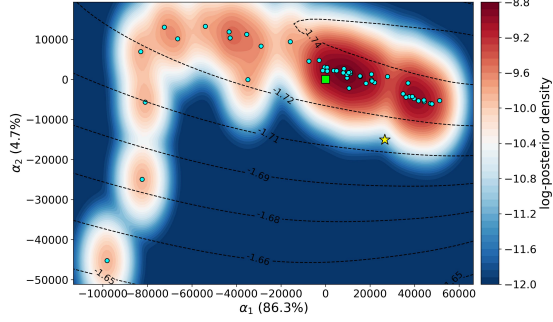


Figure 4: The PMPD plot corresponding to data lacking sub 5 Hz frequencies centered around the sample mean (green square) for Experiment 1b. The posterior samples are shown as cyan circles, and the ground truth model is a yellow star. The black dashed lines show the log-misfit contours. The percent variation of the samples along each principal component (α_1 and α_2) are shown in parenthesis on the axes.

poor generalizability of DPS to OOD models. Comparing Figures 2 and 6 we find that for OOD models the misfit minima do not overlap with the posterior modes, suggesting that even with full-bandwidth data the likelihood score is dominated by the prior score, guiding the sampler towards in-distribution models.

CONCLUSION

The PMPD plots presented in this work provide a qualitative diagnostic to assess the effectiveness of diffusion-based methods for solving seismic inverse problems. Nonetheless, we cannot definitively conclude that multimodality in this projected 2D space implies multimodality in the higher dimension space. We analyze the diffusion method in 2 special cases: (1) when data is lacking low-frequency content, and (2) when the test model is OOD relative to the training data. Experiment 1 reveals that while DPS enables high-fidelity subsurface reconstruction under ideal inversion conditions, band-limited data lacking low-frequency content still leads to cycle-skipping. The generated samples exhibit high reconstruction error with high certainty on average. Figure 4 shows the posterior mode bias toward local minima and displacement of the ground-truth from the mode. These features characterize cycle-skipping in DPS-based inversion. Experiment 2 demonstrates the poor reconstruction quality of DPS when the ground-truth is OOD. From Figure 6 we observe poor concentration of the samples, suggesting high estimation uncertainty. We also find significant separation between the ground-truth and the posterior support. These features indicate poor generalization over OOD subsurface models and tendency of the diffusion prior to drive the sampling process to in-distribution models. Cycle-skipping mitigation techniques such as low-frequency extrapolation and techniques to improve generalizability of data-driven inversion such as training data augmentation may help address these limitations.

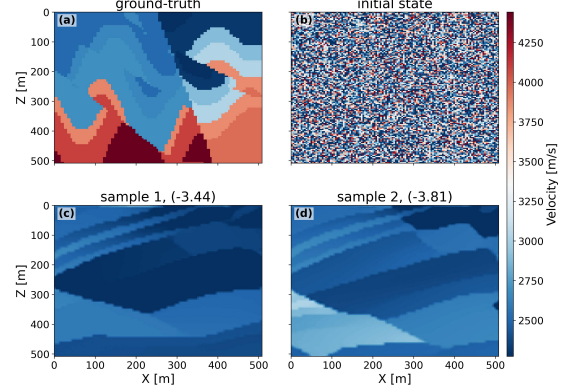


Figure 5: Experiment 2 plots: (a) The ground truth model for Experiment 2. (b) An example of the pure noise initial state. (c,d) Two (out of 50) samples generated by DPS-based inversion. The log-misfit of the posterior samples are shown in parenthesis next to the sample number.

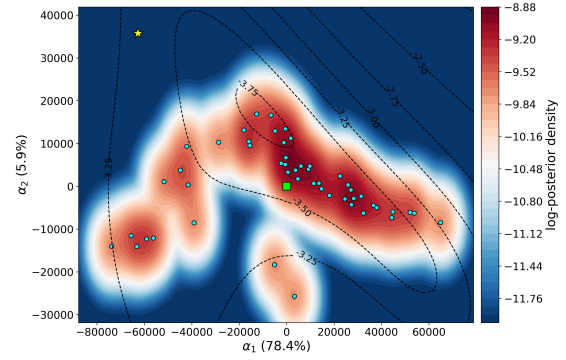


Figure 6: PMPD plot corresponding to OOD ground-truth centered around the sample mean (green square) for Experiment 2. Posterior samples are shown as cyan circles and the ground truth model is a yellow star. The black dashed lines show the log-misfit contours. The percent variation of the samples along each principal component (α_1 and α_2) are shown in parenthesis on the axes.

ACKNOWLEDGMENTS

R. Harikumar and S. Minkoff would like to acknowledge support from the sponsors of the UT Dallas 3D+4D Seismic FWI Research Consortium. Brookhaven National Laboratory is supported by the U.S. Department of Energy's Office of Science under Contract No. DE-SC0012704. We thank Dr. Bill Symes for helpful discussions on interpreting the posterior landscapes.